GUARDX SMART SECURITY SYSTEM

A MINI – PROJECT REPORT

Submitted by

DHARSHAN C	**511323106013**
PRAVEEN KUMAR N	**511323106038**
SRIDHAR SRINIVASAN	**511323106047**

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ANNA UNIVERSITY: CHENNAI 600 025

BONAFIDE CERTIFICATE

Certified that this project report **"GUARDX SMART SECURITY SYSTEM"** is the Bonafide work of **"DHARSHAN C (511323106013), PRAVEEN KUMAR N (511323106038), SRIDHAR SRINIVASAN (511323106047) "** who carried out the project work under my supervision.

SIGNATURE
Dr. M RATHIKA, M.E., Ph.D.

HEAD OF THE DEPARTMENT
Department of Electronics and Communication Engineering
Kingston Engineering College,
Chittoor main road, Vellore-632059

SIGNATURE
Mr. J. PAUL RICHARDSON GNANARAJ M.E.

ASSISTANT PROFESSOR
Department of Electronics and Communication Engineering
Kingston Engineering College,
Chittoor main road, Vellore-632059

CERTIFICATE OF EVALUATION

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S. No	NAME	PROJECT TITLE	SUPERVISOR NAME
1.	DHARSHAN C (511323106013)	GUARDX SMART SECURITY SYSTEM	Mr. J. PAUL RICHARDSON GNANARAJ, M. E
2.	PRAVEEN KUMAR N (511323106038)		
3.	SRIDHAR SRINIVASAN (511323106047)		

The report of the project work submitted by the above students in partial fulfilment for the award of **BACHELOR OF ENGINEERING DEGREE IN ELECTRONICS AND COMMUNICATION OF ANNA UNIVERSITY** was evaluated and confirmed.

The project report was submitted for viva voce held on _____, at Kingston Engineering College

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

Security systems play an important role in protecting homes, offices, and other facilities from unauthorized access. Traditional locking systems based on mechanical keys have several limitations such as key duplication, loss of keys, and lack of monitoring capabilities. To overcome these limitations, a smart lock security system using embedded technology is proposed in this project. The system is designed using an Arduino Uno microcontroller, a 4×4 keypad for password entry, and an LCD display for user interaction. The locking mechanism is controlled using a relay connected to a solenoid lock. The system implements a dual password authentication mechanism consisting of a general password and a master password. Under normal conditions, the user enters the general password to unlock the door.If the password is entered incorrectly three consecutive times, the system automatically switches to master mode. In master mode, only the master password can unlock the system. At the same time, an ESP32-CAM module is activated to provide video monitoring of the entrance area. The camera captures and transmits video, allowing the user to monitor suspicious activity. This feature enhances the security system by providing surveillance similar to a CCTV system. The proposed smart lock system improves security by combining password authentication with camera monitoring and alert mechanisms. The system is cost-effective, easy to implement, and suitable for residential and small-scale security applications.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The Internet of Things (IoT) is a modern technology that enables the connection of computing devices and everyday objects through the internet without requiring direct human interaction. IoT systems consist of smart devices equipped with processors, sensors, and communication modules that can collect, transmit, and process data from their surroundings. This data is either analyzed locally or sent to cloud platforms through gateways, enabling real-time monitoring and control. As a key component of Industry 4.0, IoT integrates the physical and digital worlds, providing flexible connectivity and advanced automation capabilities.

Doors play a crucial role in ensuring the safety and privacy of homes, offices, and other facilities. They serve as the primary access point for entry and exit, making them an essential component of any security system. Traditionally, mechanical locks have been widely used for securing doors. However, these locks have limitations such as the risk of key loss, duplication, and lack of access control, which can compromise security.

With rapid advancements in technology, door security systems have evolved significantly. Modern digital lock systems provide more secure and efficient solutions compared to traditional locks. These systems allow access through various methods such as passwords, fingerprint recognition, facial recognition, and smart cards. Such technologies enhance both convenience and security by eliminating the need for physical keys and enabling better access management.

Several research works have been carried out to improve door security systems. For instance, fingerprint-based systems using microcontrollers like Arduino have been developed, where registered fingerprints are used to unlock doors through a solenoid mechanism. Similarly, IoT-based smart lock systems using ESP32 microcontrollers have been designed to provide remote access and monitoring through mobile applications. Keypad-based digital lock systems using PIN authentication have also been implemented to enhance security and usability.

The major advantage of modern smart lock systems is their ability to precisely control and monitor access. They solve common issues such as lost keys, unauthorized duplication, and the complexity of managing multiple keys. However, many existing systems still lack features such as remote monitoring, real-time alerts, and flexible multi-level authentication.

To address these limitations, this project proposes the development of a **GUARDX Smart Security System**, which integrates IoT technology with advanced authentication methods. The system provides dual-layer security using user and master PINs, automatic system lockdown after multiple failed attempts, and real-time notifications. It also enables remote monitoring and control, making it a secure, efficient, and user-friendly solution for modern access control applications.

1.2 EVOLUTION OF LOCKING SYSTEMS IN SECURITY

Locking systems have evolved over time from simple mechanical devices to advanced digital systems. The following sections describe the earlier stages and their limitations.

1.2.1 Traditional Mechanical Locks

In earlier times, doors were secured using mechanical locks and physical keys.

Explanation:

- Used metal keys for locking and unlocking
- Keys could be easily lost or duplicated
- No monitoring or access tracking
- Provided only basic security



FIG 1.1 Traditional Mechanical Locks

1.2.2 Keypad-Based Digital Locks

Keypad-based systems use PIN codes for authentication.

Explanation:

- Users enter a password to unlock the door
- More secure than traditional locks
- Easy to operate and maintain
- Vulnerable to password guessing or sharing



FIG 1.2 Keypad-Based Digital Locks

1.2.3 RFID and Card-Based Systems

RFID technology allows access using cards or tags.

Explanation:

- Contactless access system
- Faster and more convenient
- Cards can be lost or stolen
- Requires additional hardware



FIG 1.3 RFID and Card-Based Systems

1.2.4 Biometric Lock Systems

Biometric systems use fingerprint or facial recognition for authentication.

Explanation:

- High level of security
- Unique identification for each user
- Expensive compared to other systems
- Requires proper maintenance



FIG 1.4 Biometric Lock Systems

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The rapid development of Internet of Things (IoT) technology has significantly transformed modern security systems. With the increasing demand for safety, automation, and remote accessibility, traditional security mechanisms are being replaced by smart and intelligent systems. Smart door lock systems, in particular, have gained attention due to their ability to integrate embedded systems, wireless communication, and advanced authentication techniques.

IoT-based security systems enable devices to communicate with each other and share data in real time, allowing users to monitor and control access remotely. These systems enhance convenience, reduce human effort, and improve overall security. This chapter presents a detailed review of various research works related to IoT-based systems, smart home security, and digital door lock mechanisms, along with recent advancements in the field.

2.2 IOT-BASED SYSTEMS

According to Atzori et al. [1], the Internet of Things enables seamless communication between devices through interconnected networks, allowing real-time data exchange and intelligent decision-making. Their work highlights the importance of IoT in creating smart environments, including applications in healthcare, transportation, and home automation.

Madakam et al. [2] discussed the architecture and applications of IoT, emphasizing its role in automation, scalability, and remote monitoring. The study explains how IoT devices equipped with sensors and communication modules can improve efficiency and reduce human intervention in systems such as home security and industrial automation.

In recent years, IoT technology has advanced significantly. Zhang et al. [6] (2023) proposed an intelligent IoT framework for smart environments that supports real-time monitoring and predictive analysis. Their system demonstrates how IoT can improve system reliability and responsiveness in security applications.

Furthermore, Kumar et al. [7] (2024) developed an IoT-enabled smart monitoring system using cloud integration. Their work highlights the importance of cloud computing in storing and analyzing large amounts of data generated by IoT devices, enabling better decision-making and system optimization.

2.3 SMART HOME SECURITY SYSTEMS

Alaa et al. [3] reviewed various smart home security systems based on IoT technology. The study explains how integrating sensors, cameras, and communication modules enhances home security. Features such as real-time alerts, remote monitoring, and automation are identified as key components of modern smart homes.

Recent studies have focused on improving the intelligence and efficiency of these systems. Sharma and Singh [8] (2023) developed a smart home security system that integrates motion detection sensors and mobile notifications. Their system alerts users instantly in case of unauthorized access, improving response time.

Patel et al. [9] (2024) proposed an AI-integrated smart home security system that uses machine learning algorithms to detect unusual activities. This system enhances security by identifying suspicious patterns and reducing false alarms.

In addition, Rahman et al. [10] (2025) introduced a cloud-based smart security system that provides real-time video streaming and remote access control. Their system demonstrates improved scalability and reliability for large-scale deployments.

2.4 DIGITAL DOOR LOCK SYSTEMS

Kumar and Sowmya [4] developed an Arduino-based smart door lock system using RFID and password authentication. Their system allows access only to authorized users and improves security compared to traditional locks. However, the system has limitations such as lack of remote monitoring and dependency on physical RFID cards.

Murthy [5] proposed an IoT-based smart security and home automation system that integrates multiple devices for remote control and monitoring. The system allows users to manage access and monitor security through mobile applications, enhancing convenience and efficiency.

Recent advancements have further improved digital lock systems. Lee et al. [11] (2023) designed a smart door lock system using Bluetooth and mobile applications, enabling secure and easy access control. However, the system has limited range due to Bluetooth constraints.

Gupta et al. [12] (2024) developed a multi-authentication smart lock system using fingerprint and PIN-based access. Their system improves security by combining two authentication methods but lacks real-time alert features.

In 2025, Ahmed et al. [13] proposed an IoT-based smart lock system using ESP32 with cloud integration and mobile application control. The system provides real-time notifications, remote access, and improved security features, making it more suitable for modern applications.

2.5 RESEARCH GAP

From the above studies, it is observed that significant advancements have been made in smart lock and IoT-based security systems. However, several limitations still exist:

- Most systems rely on a single authentication method such as RFID, password, or fingerprint
- Limited implementation of multi-layer security mechanisms
- Lack of real-time alert and notification systems in some designs
- Insufficient focus on intrusion detection and automatic system response
- Limited integration with cloud and mobile-based monitoring in older systems

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING LOCKING MECHANISMS

Traditional security systems mainly rely on mechanical or basic electronic locking mechanisms. These systems are commonly used in homes, offices, and commercial buildings. Although they provide a basic level of protection, they often lack advanced security features such as intelligent monitoring, multi-level authentication, and real-time alerts. Some of the commonly used existing systems are mechanical locks, RFID-based systems, and biometric security systems.

3.1.1 Mechanical Key-Based Locks

Mechanical locks are the most widely used locking systems in residential and commercial buildings.These locks operate using a physical key that fits into the lock mechanism. When the correct key is inserted and turned, the lock opens. Although these locks are simple and inexpensive, they have several security weaknesses. Keys can be lost, duplicated, or stolen, which may allow unauthorized access. Mechanical locks also do not provide any alert or monitoring capability if someone attempts to break or tamper with the lock.



FIG 3.1 Mechanical Key-Based Locks

3.1.2 RFID-Based Access Systems

Radio Frequency Identification (RFID) based locking systems use RFID cards or tags to authenticate users. When a user places the RFID card near the reader, the system verifies the card ID and allows access if the card is authorized. These systems are commonly used in offices, hotels, and institutional buildings. However, RFID cards can be lost or stolen, and unauthorized users may gain access if they obtain the card.



FIG 3.2 RFID-Based Access Systems

3.1.3 Biometric Lock Systems

Biometric locking systems use biological characteristics such as fingerprints, face recognition, or iris patterns to identify authorized users. These systems offer a higher level of security compared to mechanical locks and RFID systems. However, biometric systems are more expensive and require specialized hardware. Environmental factors such as dust, moisture, or lighting conditions may also affect their performance.



FIG 3.3 Biometric Lock Systems

3.1.4 Face Recognition-Based Systems

Face recognition systems use camera modules and image processing techniques to identify users based on their facial features. These systems provide contactless authentication and improved security.

However, their performance depends on lighting conditions, camera quality, and processing capability. In low-light environments or with changes in appearance, recognition accuracy may decrease. Also, many existing systems do not provide backup authentication methods if face recognition fails.



FIG 3.4 Face Recognition-Based Systems

3.1.5 Basic IoT-Based Lock Systems

Basic IoT-based lock systems use Wi-Fi or Bluetooth modules to enable remote access and control through mobile applications. These systems allow users to lock or unlock doors from anywhere and receive basic notifications.

However, many existing IoT systems lack strong security mechanisms, such as multi-layer authentication and intrusion detection. They may also be vulnerable to network-related issues and unauthorized access if not properly secured.



FIG 3.5 Basic IoT-Based Lock Systems

3.2 DISADVANTAGES OF EXISTING SYSTEMS

Despite their widespread use, existing locking systems have several limitations:

- **Physical Vulnerability:** Mechanical locks rely entirely on physical keys, which can be easily duplicated or misplaced.
- **Loss of Tokens:** RFID systems depend on cards or tags that can be lost or stolen.
- **High Cost and Complexity:** Biometric and face recognition systems require expensive hardware and complex implementation.
- **Environmental Dependency:** Biometric and face recognition systems may fail due to lighting, dust, or environmental conditions.
- **Limited Security Layers:** Most systems use only one authentication method, reducing overall security.
- **No Active Monitoring:** Many traditional systems do not provide real-time alerts or notifications during unauthorized access attempts.
- **Weak IoT Security:** Basic IoT systems may lack proper encryption and secure communication, making them vulnerable to cyber threats.

CHAPTER 4

PROPOSED SYSTEM

4.1 ARCHICTECTURE OF PROPOSED SYSTEM

The proposed system is a smart door lock security system designed using an Arduino Uno microcontroller and an ESP32-CAM module. The system uses a 4×4 keypad for password input and an LCD display to provide user instructions and status messages. A solenoid lock controlled through a relay is used as the locking mechanism. The system implements a dual password authentication mechanism consisting of a general password and a master password.

Under normal conditions, the user enters the general password using the keypad. If the password is correct, the Arduino activates the relay and unlocks the door. If the password is entered incorrectly three times, the system switches to master mode. In master mode, only the master password can unlock the system. At the same time, the ESP32-CAM module is activated to monitor the entrance area and send video alerts to the user. This feature acts as a surveillance system similar to a CCTV camera and enhances the security of the system.

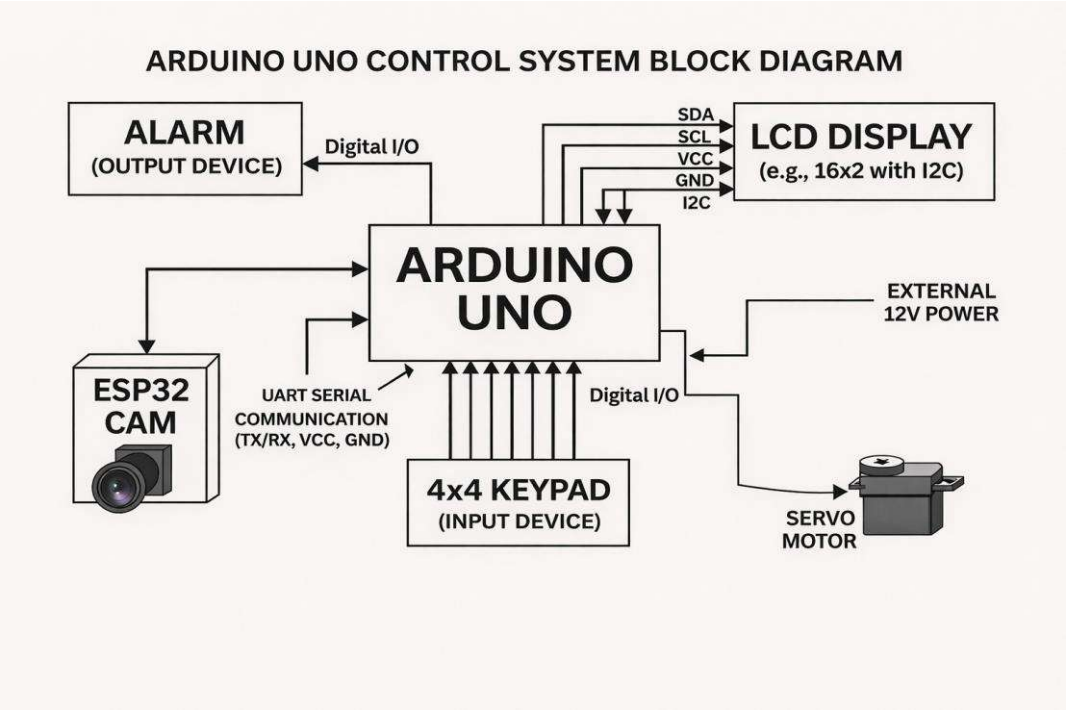


FIG 3.1 ARCHICTECTURE OF PROPOSED SYSTEM

4.2 PROPOSED SYSTEM FLOW CHART

The workflow of the *GuardX Smart Security System* represents the step-by-step operational process of the system, starting from user interaction to security response. It illustrates how the system authenticates users using dual-layer security (General PIN and Master PIN), monitors unauthorized access attempts, and performs actions such as system unlocking, alert generation, and automatic lockdown. This workflow ensures a structured and secure functioning of the system by integrating hardware components like keypad, ESP32-CAM, and solenoid lock with software-based decision making and real-time notifications. the ESP32-CAM for surveillance. The system remains in master mode until the correct master password is entered.

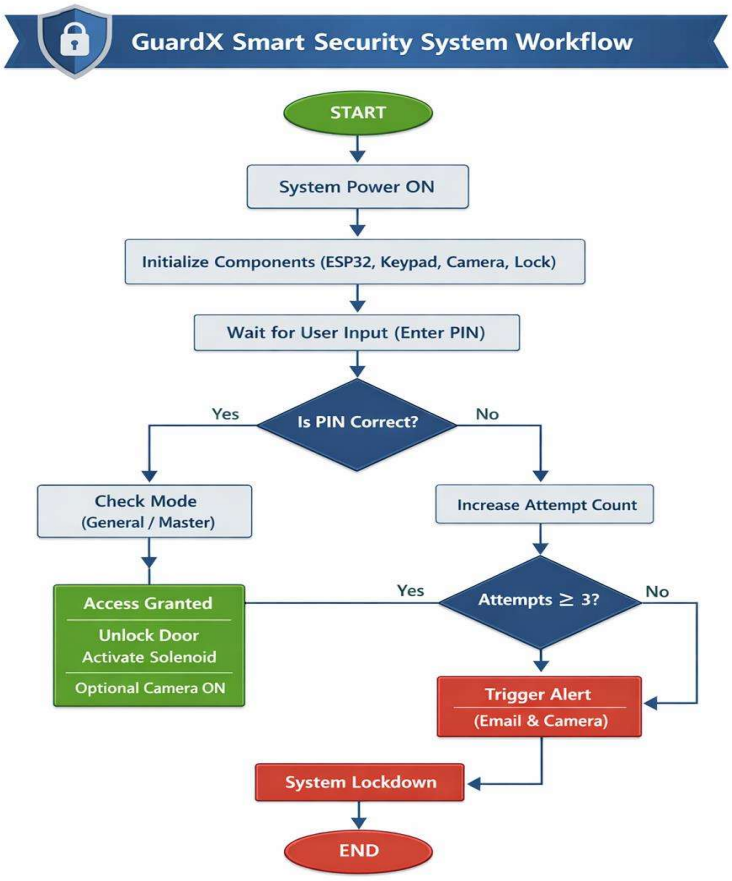


FIG 4.2 PROPOSED SYSTEM FLOW CHART

4.3 DUAL PASSWORD LOGIC IMPLEMENTATION

The system uses a dual-level authentication mechanism to increase security.

4.3.1 General Mode Operation

In general mode, the system operates under normal conditions. The user enters the general password using the keypad. The Arduino microcontroller verifies the entered password by comparing it with the stored password in the program. If the password matches, the system activates the relay and unlocks the solenoid lock, allowing access. The LCD display shows a message indicating that the door has been unlocked.

4.3.2 Master Mode Operation

If the user enters an incorrect password three consecutive times, the system automatically switches to master mode. In this mode, the general password is temporarily disabled, and only the master password can unlock the system. This prevents unauthorized users from repeatedly attempting passwords. The master password acts as a secure administrative password known only to the system owner.

4.3.3 Alert and Camera Activation Mechanism

When the system enters master mode due to multiple incorrect password attempts, the Arduino sends a signal to the ESP32-CAM module. The ESP32-CAM activates its camera and begins monitoring the entrance area. The captured video can be streamed over a network and viewed by the user. This feature helps the user identify suspicious activity and provides additional security by recording unauthorized access attempts.

4.4 KEY FEATURES AND BENEFITS OF THE PROPOSED SYSTEM

The proposed **GUARDX Smart Security System** provides enhanced security using a **dual password authentication mechanism** with user and master access levels. This improves protection by adding an extra layer of verification.

The system performs **automatic password verification** and detects multiple incorrect attempts. In such cases, it activates a **security response like system lockdown**, preventing unauthorized access.

An **ESP32-CAM module** is integrated to provide surveillance. It captures images or streams video when suspicious activity is detected, allowing users to monitor the entrance remotely.

The system also includes **IoT connectivity**, enabling users to receive real-time alerts and monitor the system through mobile devices. Additionally, **automatic locking** ensures the door is secured after access, reducing the risk of leaving it open.

4.5 ADVANTAGES OF THE PROPOSED SYSTEM

- **High Security:** Dual authentication reduces unauthorized access
- **Real-Time Monitoring:** Live alerts and remote access through IoT
- **No Physical Keys Required:** Eliminates key loss and duplication issues
- **Cost-Effective:** Uses affordable components like Arduino and ESP32-CAM
- **Automatic Locking:** Ensures door is always secured
- **Intrusion Detection:** Detects multiple failed attempts and responds
- **Easy to Use:** Simple operation and user-friendly system
- **Scalable:** Can be upgraded with additional features in future

CHAPTER 5

TOOLS REQUIRED

5.1 TOOLS AND TECHNOLOGIES EMPLOYED

The development of the smart lock security system involves the use of both hardware and software tools to design, program, and test the overall system effectively. The hardware components include an **Arduino Uno microcontroller**, which acts as the main control unit, an **ESP32-CAM module** for surveillance and image capture, a **4x4 matrix keypad** for password input, and a **16x2 LCD display** to show system messages and status. Additionally, a **servo motor** are used to control the locking mechanism, while a **buzzer** is included to provide alert signals during unauthorized access attempts. On the software side, the system is developed using the **Arduino Integrated Development Environment (IDE)**, which is used to write, compile, and upload the program to the Arduino Uno. The combination of these hardware and software tools enables the system to perform secure password authentication, operate the locking mechanism efficiently, and activate the camera-based surveillance system when suspicious activity is detected.

5.2 HARDWARE TOOLS

The development of the smart lock security system uses both hardware and software components. The main hardware component is the **Arduino Uno microcontroller**, which controls the entire system. A **4x4 keypad** is used for password input, and a **16x2 LCD display** shows system messages. The **ESP32-CAM module** provides surveillance, while a **servo motor** and **relay module** control the locking mechanism. A **buzzer** is used for alerts during unauthorized access.

The system is programmed using the **Arduino IDE**, which allows code writing, compiling, and uploading to the Arduino Uno. These tools work together to create a simple, cost-effective, and secure smart lock system.

5.2.1 ARDUINO UNO

The first step in developing the system is installing the **Arduino IDE** software on a computer. The Arduino IDE is used to write, compile, and upload programs to the **Arduino Uno** microcontroller.

After installation, the Arduino Uno board is connected to the computer using a USB cable. The correct board (**Arduino Uno**) and COM port are selected in the Arduino IDE to establish proper communication.

The required libraries for components such as the **keypad** and **LCD display** are then installed in the Arduino IDE. These libraries help in easily interfacing the hardware components with the Arduino Uno.

Once the setup is completed, the system is ready for coding and uploading the program to the Arduino Uno for further implementation.

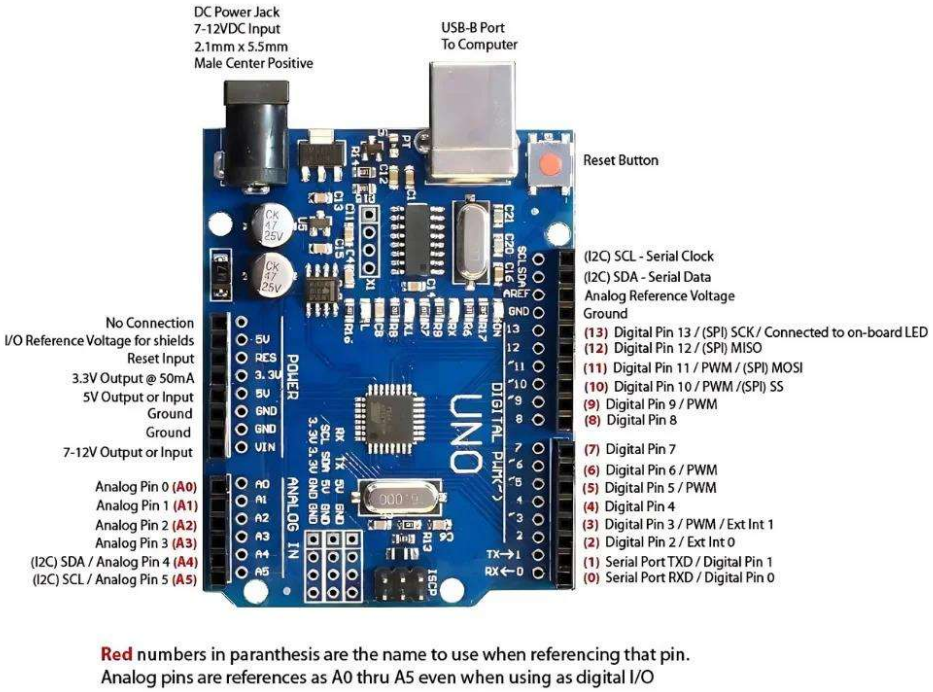


FIG 5.1 ARDUINO UNO

5.2.2 ESP32-CAM Setup

The ESP32-CAM module is used to provide camera monitoring. It is configured to capture images or video when the Arduino switches the system to master mode. The camera allows the user to monitor the entrance area and detect unauthorized access attempts.

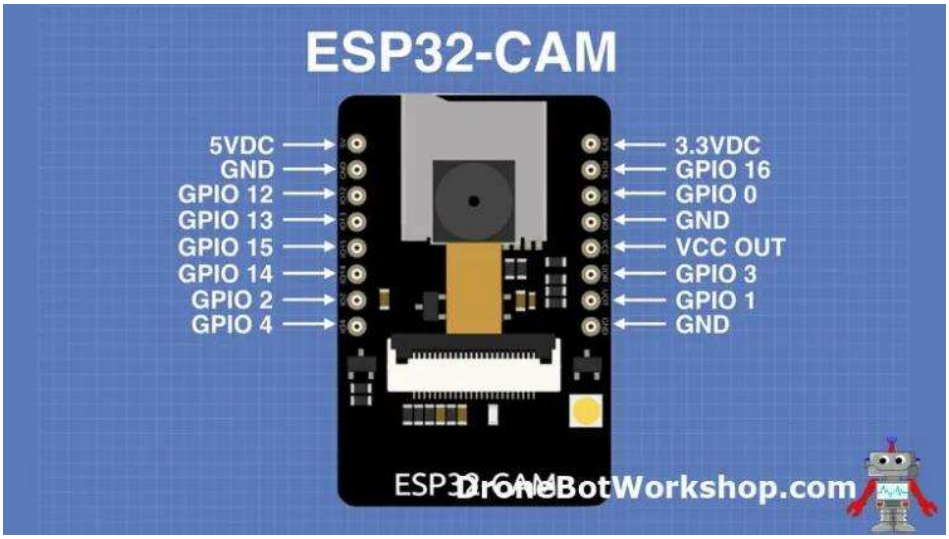


FIG 3.2 ESP 32 – CAM

5.2.3 Keypad Interfacing

The 4×4 keypad is connected to the digital input pins of the Arduino Uno. It allows the user to enter the password for authentication. The standard keypad library in Arduino is used to detect which key is pressed.



FIG 5.3 4X4 Keypad

5.2.4 LCD Integration

The LCD display is used to show system messages such as “Enter Password,” “Access Granted,” or “Wrong Password.” It is connected to the Arduino using digital pins or an I2C interface to save wiring.



FIG 5.4 LCD Didplay

5.2.5 Lock Control Using Servo Motor

The servo motor is used as the locking mechanism in place of a solenoid lock. Unlike solenoid locks that require higher power and a relay module, a servo motor can be directly controlled by the Arduino using PWM (Pulse Width Modulation) signals.When the correct password is entered, the Arduino sends a signal to the servo motor to rotate to a specific angle (for example, 90°), which unlocks the door. When the system is locked, the servo motor returns to its original position (0°), securing the door.



FIG 5.5 Servo Motor

5.2.6 Buzzer Alert System

The buzzer is connected to one of the Arduino output pins. It produces a sound when an incorrect password is entered or when the system detects suspicious activity.



FIG 5.6 Buzzer

5.2.7 Power Supply

The smart lock system is powered using either a **battery** or a **12V DC adapter**. The Arduino Uno regulates the voltage to supply required power to all components. A stable power source ensures reliable and smooth system operation.



FIG 5.7 Power Supply

5.3 SOFTWARE TOOL

5.3.1ARDUINO IDE

The **Arduino Integrated Development Environment (IDE)** is the primary software tool used for programming the smart lock system. It provides a user-friendly platform to write, compile, and upload code to the Arduino Uno microcontroller. The programming language used in the Arduino IDE is based on C and C++, making it easy to understand and implement. The IDE also includes built-in libraries that simplify interfacing with components such as the keypad, LCD display, and other modules. Additionally, it offers a serial monitor feature that helps in debugging and monitoring the system output in real time. Overall, the Arduino IDE plays a crucial role in developing and testing the functionality of the smart lock system.

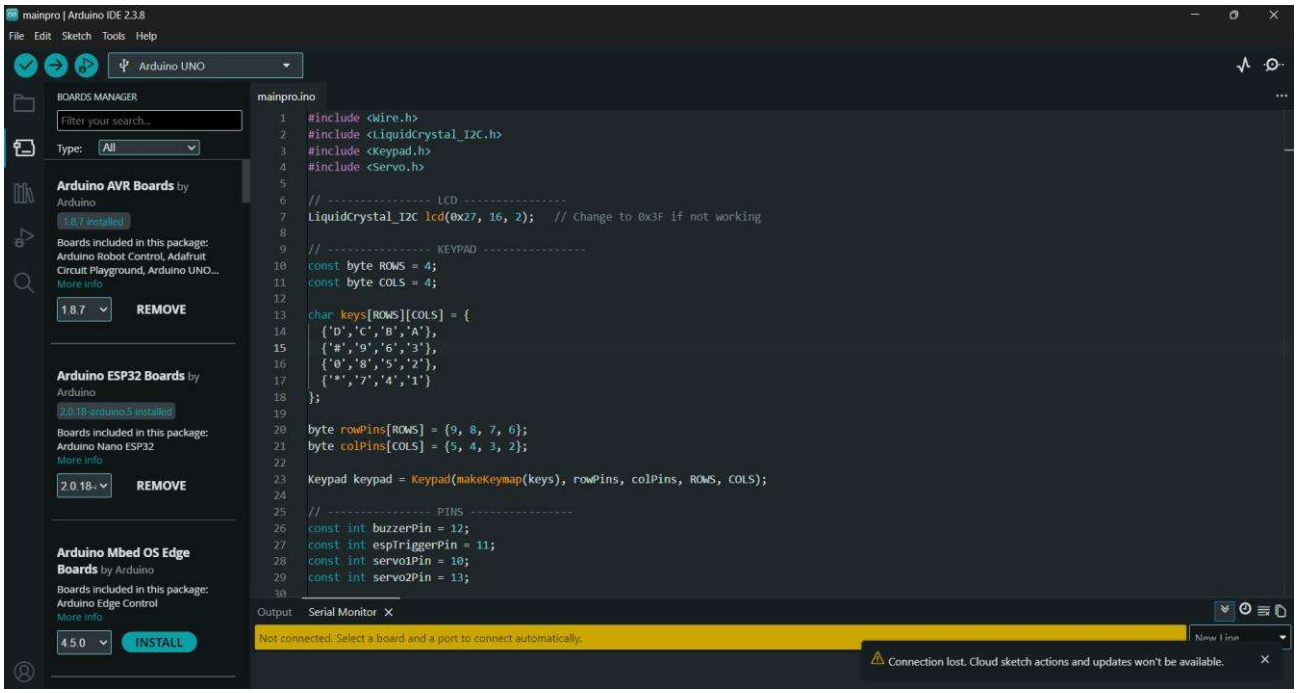


FIG 5.8 Arduino IDE 2.3.8

CHAPTER 6

RESULTS AND DISCUSSION

The results of the smart lock security system demonstrate that the components and microcontroller operate effectively and reliably. The **keypad module** successfully detects user input when a password is entered and sends a signal to the **Arduino Uno microcontroller**.

Based on this input, the Arduino verifies the password. When the correct password is entered, the **servo motor** automatically unlocks the door, enhancing user convenience and security. The **LCD display** shows messages such as "ACCESS GRANTED" or "ACCESS DENIED," providing clear feedback to the user.

If an incorrect password is entered, the system activates the **buzzer** to alert the user. After multiple incorrect attempts, the system initiates a **security response**, such as temporary system lock, to prevent unauthorized access.

The **ESP32-CAM module** also performs effectively by capturing images or enabling monitoring when suspicious activity is detected. This allows the user to observe the entrance area remotely.

The Arduino processes all inputs accurately and controls the servo motor and other components accordingly. The system performs authentication, alerting, and locking operations efficiently under real-time conditions.

Overall, the performance of the system is consistent and reliable, demonstrating that the proposed design can significantly improve modern security systems.

6.1 IMPLEMENTATION OF PROPOSED SYSTEM

The results obtained from the smart lock system show that all components function correctly. The **keypad** successfully detects the password input and sends the data to the Arduino Uno microcontroller.

The Arduino compares the entered password with the stored password. If the password is correct, the system grants access by activating the servo motor. If the password is incorrect, access is denied and the buzzer is triggered as an alert.

The **ESP32-CAM module** is activated during multiple failed attempts or suspicious activity. It captures images or enables monitoring, enhancing the security level of the system.

The system operates efficiently with accurate input detection and quick response. The results confirm that the system is capable of performing secure authentication and monitoring effectively.

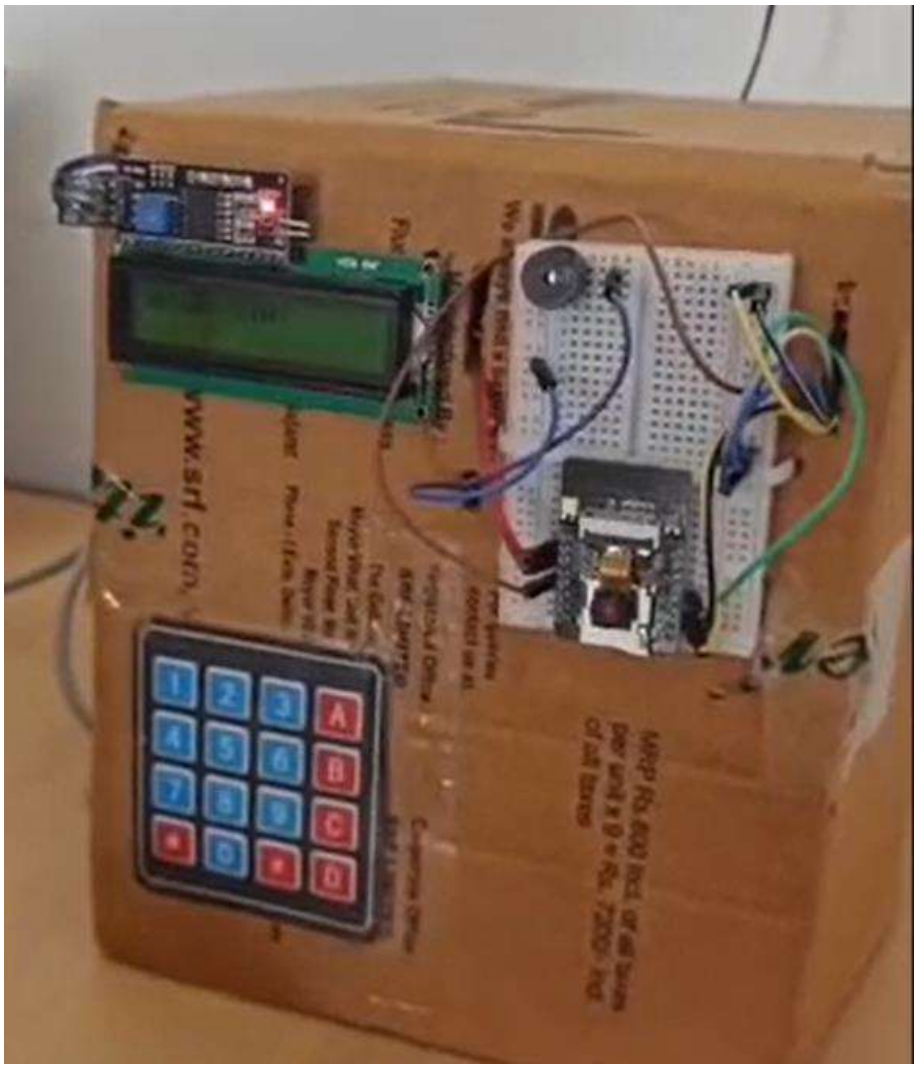


FIG 6.1 Integration of Components with Arduino Uno microcontroller.

6.2 WORKING OF PROPOSED SYSTEM

The **servo motor** plays an important role in the operation of the smart lock system. When the correct password is entered, the Arduino sends a signal to the servo motor to unlock the door. The locking mechanism operates smoothly, allowing access.

After a certain period, the door can automatically lock again to ensure safety. In case of incorrect attempts, the servo motor remains inactive, preventing unauthorized access.

The system also integrates the **ESP32-CAM**, which activates during suspicious conditions to capture images or provide monitoring. The response of all components is quick and accurate, ensuring smooth and reliable system operation.

The successful actuation of the servo motor and proper functioning of all modules confirm that the smart lock system works effectively

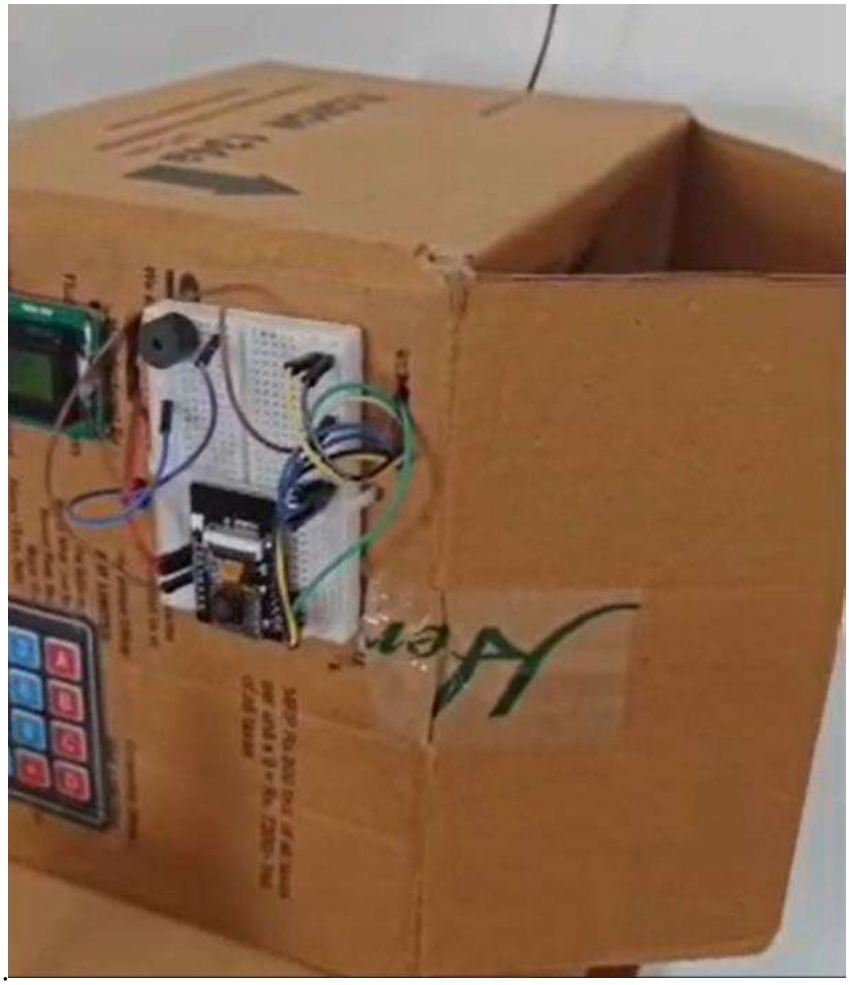


FIG 6.2 Servo Motor Working



FIG 6.3 SECUITY MAIL when SYSTEM at MASTER MODE



FIG 6.4 SYSTEM RESET MAIL at GENERAL MODE

FIG 6.5 ESP32-CAM LIVE MONITORING

6.3 ANALYSIS OF EXISTING VS PROPOSED SYSTEM FUNCTIONALITIES

The comparison between existing locking systems and the proposed smart lock system highlights significant improvements in terms of security, automation, and user convenience. Traditional locking systems mainly rely on mechanical keys or basic electronic methods, which provide limited security and lack monitoring capabilities. These systems do not offer real-time alerts or multi-level authentication, making them vulnerable to unauthorized access.

In contrast, the proposed **GUARDX Smart Security System** uses advanced technologies such as keypad-based authentication, IoT connectivity, and camera surveillance. It provides dual-layer password protection, automatic locking, and intrusion detection features. The integration of the **ESP32-CAM module** allows real-time monitoring, while the Arduino Uno ensures accurate control of all components. Overall, the proposed system offers enhanced security, better user experience, and flexibility for future upgrades compared to existing systems.

S.No	Parameter	Existing System (Traditional Locks)	Proposed System (Smart Lock System)
1	Power Supply	No power required (manual operation)	Uses Arduino power supply (USB / 5V–12V)
2	Functionality	Manual locking and unlocking	Automated password-based locking system
3	Input Devices	Physical key	Keypad for password input
4	Output Devices	No output indication	LCD display, buzzer, servo motor
5	Communication	No communication system	IoT-based monitoring and alerts
6	Security Features	Basic security	Dual password, intrusion detection
7	Data Storage	No storage	Stores password data in microcontroller
8	Target Application	Homes and basic use	Homes, offices, smart security systems
9	Scalability	Cannot be upgraded	Can be extended with IoT, camera, biometrics
10	Energy Efficiency	No energy usage	Low power consumption system
11	Accessibility	Requires physical key	Keypad and remote access
12	Monitoring	No monitoring	Real-time monitoring using ESP32-CAM
13	Ease of Use	Simple but less secure	Easy to use, secure, and user-friendly

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

The smart lock security system developed in this project provides an efficient and reliable solution for modern access control. The system uses a **keypad, microcontroller, and servo motor** to perform secure authentication and control the locking mechanism. The **keypad module** allows users to enter a password, and the **Arduino Uno microcontroller** verifies the input and controls the system accordingly.

When the correct password is entered, the system unlocks the door, while incorrect attempts trigger alerts using the **buzzer**. The integration of the **ESP32-CAM module** enhances security by providing surveillance and monitoring capabilities during suspicious activity. The **LCD display** provides clear status messages, improving user interaction.

The system reduces the dependency on physical keys, thereby eliminating issues such as key loss or duplication. It also improves security through features like multiple attempt detection and automatic system response. Overall, the smart lock system is a **cost-effective, user-friendly, and secure solution** suitable for homes and small office applications.

7.2 FUTURE WORK

The smart lock system can be further enhanced by integrating advanced technologies. In the future, **biometric authentication methods** such as fingerprint or facial recognition can be added to increase the level of security.

The system can also be improved by developing a **dedicated mobile application** for better remote control and monitoring. Integration with cloud platforms can allow secure data storage and access from anywhere.

Additionally, **voice control and smart assistant integration** can be implemented for improved user convenience. The use of **advanced encryption techniques** can further enhance data security in IoT communication.

Power efficiency can also be improved by incorporating **battery backup or solar power systems**, ensuring continuous operation during power failures. These improvements can make the system more intelligent, secure, and suitable for large-scale applications.

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This paper explores edge computing to process security data locally. It reduces latency and improves real-time response in smart systems.

APPENDIX

ARDUINO IDE

ARDUINO UNO CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Keypad.h>
#include <Servo.h>

// ---------------- LCD ----------------
LiquidCrystal_I2C lcd(0x27, 16, 2);

// ---------------- KEYPAD ----------------
const byte ROWS = 4;
const byte COLS = 4;

char keys[ROWS][COLS] = {
  {'D','C','B','A'},
  {'#','9','6','3'},
  {'0','8','5','2'},
  {'*','7','4','1'}
};

byte rowPins[ROWS] = {9, 8, 7, 6};
byte colPins[COLS] = {5, 4, 3, 2};

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

// ---------------- PINS ----------------
const int buzzerPin = 12;
const int espTriggerPin = 11;
const int servo1Pin = 10;
const int servo2Pin = A0;   // ☑ FIXED (NOT 13)

// ---------------- OBJECTS ----------------
Servo servo1;
Servo servo2;

// ---------------- VARIABLES ----------------
String inputPassword = "";
```

```
const String GENERAL_PIN = "1234";
const String MASTER_PIN  = "5678";

int fails = 0;
bool masterMode = false;

// ---------------- SETUP ----------------
void setup() {
  lcd.init();
  lcd.backlight();

  pinMode(buzzerPin, OUTPUT);
  pinMode(espTriggerPin, OUTPUT);

  servo1.attach(servo1Pin);
  servo2.attach(servo2Pin);

  // LOCK POSITION
  servo1.write(90);
  servo2.write(60);

  digitalWrite(espTriggerPin, LOW);

  showMsg("SYSTEM READY", "ENTER PIN:");
}

// ---------------- LOOP ----------------
void loop() {

  // Alarm beep in master mode
  if (masterMode) {
    tone(buzzerPin, 1000, 200);
    delay(150);
  } else {
    noTone(buzzerPin);
  }

  char key = keypad.getKey();

  if (key) {
    tone(buzzerPin, 2000, 50);
    delay(150);

    if (key == 'A') {
      if (inputPassword.length() > 0) checkPass();
    }
```

```
    else if (key == 'B') {
      inputPassword = "";
      showMsg(masterMode ? "MASTER MODE" : "ENTER PIN:", "");
    }

    else if ((key >= '0' && key <= '9') || key == '#') {
      if (inputPassword.length() < 16) {
        inputPassword += key;
        lcd.setCursor(inputPassword.length() - 1, 1);
        lcd.print("*");
      }
    }
  }
}

// ---------------- PASSWORD CHECK ----------------
void checkPass() {

  if (!masterMode) {

    if (inputPassword == GENERAL_PIN) {
      showMsg("ACCESS GRANTED", "OPENING DOOR");

      // OPEN
      servo1.write(180);
      servo2.write(0);

      delay(5000);

      // CLOSE
      servo1.write(90);
      servo2.write(60);

      fails = 0;
    }

    else {
      fails++;

      int triesLeft = max(0, 3 - fails);
      showMsg("WRONG PIN", "TRY: " + String(triesLeft));

      delay(1500);

      if (fails >= 3) {
```

```
        masterMode = true;

        // ESP Trigger
        digitalWrite(espTriggerPin, HIGH);
        delay(300);
        digitalWrite(espTriggerPin, LOW);

        showMsg("SYSTEM LOCKED", "ALARM ACTIVE");
      }
    }
  }

  else {
    if (inputPassword == MASTER_PIN) {
      masterMode = false;

      digitalWrite(espTriggerPin, LOW);

      showMsg("DISARMED", "SYSTEM RESET");

      fails = 0;
      delay(2000);
    }

    else {
      showMsg("WRONG MASTER", "TRY AGAIN");
      delay(2000);
    }
  }

  inputPassword = "";
  showMsg(masterMode ? "MASTER MODE" : "ENTER PIN:", "");
}

// ---------------- DISPLAY FUNCTION ----------------
void showMsg(String l1, String l2) {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(l1);
  lcd.setCursor(0, 1);
  lcd.print(l2);
}
```

ESP32 – CAM CODE:

```
#include "esp_camera.h"
#include <WiFi.h>
#include "esp_http_server.h"
#include <ESP_Mail_Client.h>

// ===========================
// WIFI CONFIG
// ===========================
const char* ssid = "OOADLAB";
const char* password = "kingston123";

// ===========================
// EMAIL CONFIG
// ===========================
#define SMTP_HOST "smtp.gmail.com"
#define SMTP_PORT 465

#define AUTHOR_EMAIL "praveenkumar20064100@gmail.com"
#define AUTHOR_PASSWORD "nnfp kqdm hjpv sdex"   // App Password

#define RECIPIENT_EMAIL_1 "dharshan93648@gmail.com"
#define RECIPIENT_EMAIL_2 "praveenkumar20064100@gmail.com"

// ===========================
// TRIGGER PIN (FIXED)
// ===========================
#define TRIGGER_PIN 14   // ☑ Changed from 13

// ===========================
// CAMERA PINS (AI THINKER)
// ===========================
#define PWDN_GPIO_NUM  32
#define RESET_GPIO_NUM -1
#define XCLK_GPIO_NUM   0
#define SIOD_GPIO_NUM  26
#define SIOC_GPIO_NUM  27
#define Y9_GPIO_NUM    35
#define Y8_GPIO_NUM    34
#define Y7_GPIO_NUM    39
#define Y6_GPIO_NUM    36
#define Y5_GPIO_NUM    21
#define Y4_GPIO_NUM    19
#define Y3_GPIO_NUM    18
#define Y2_GPIO_NUM     5
#define VSYNC_GPIO_NUM 25
```

```
#define HREF_GPIO_NUM  23
#define PCLK_GPIO_NUM  22

// ===========================
SMTPSession smtp;
httpd_handle_t stream_httpd = NULL;

bool isLocked = false;

// ===========================
// STREAM FUNCTION
// ===========================
static esp_err_t stream_handler(httpd_req_t *req){
  camera_fb_t * fb = NULL;
  esp_err_t res = ESP_OK;
  char part_buf[64];

  httpd_resp_set_type(req, "multipart/x-mixed-replace; boundary=frame");

  while(true){
    fb = esp_camera_fb_get();
    if (!fb) return ESP_FAIL;

    size_t hlen = snprintf(part_buf, 64,
      "\r\n--frame\r\nContent-Type: image/jpeg\r\nContent-Length: %u\r\n\r\n",
      fb->len);

    res = httpd_resp_send_chunk(req, part_buf, hlen);

    if(res == ESP_OK){
      res = httpd_resp_send_chunk(req, (const char *)fb->buf, fb->len);
    }

    esp_camera_fb_return(fb);

    if(res != ESP_OK) break;
  }

  return res;
}

// ===========================
// START SERVER
// ===========================
void startCameraServer(){
  httpd_config_t config = HTTPD_DEFAULT_CONFIG();
```

```
  config.server_port = 80;

  httpd_uri_t stream_uri = {
    .uri = "/",
    .method = HTTP_GET,
    .handler = stream_handler,
    .user_ctx = NULL
  };

  if (httpd_start(&stream_httpd, &config) == ESP_OK) {
    httpd_register_uri_handler(stream_httpd, &stream_uri);
  }
}

// ==========================
// EMAIL FUNCTION
// ==========================
void sendStatusEmail(bool locked, String streamUrl) {

  ESP_Mail_Session session;
  session.server.host_name = SMTP_HOST;
  session.server.port = SMTP_PORT;
  session.login.email = AUTHOR_EMAIL;
  session.login.password = AUTHOR_PASSWORD;
  session.secure.mode = esp_mail_secure_mode_ssl_tls;

  SMTP_Message message;

  message.sender.name = "ESP32 Security System";
  message.sender.email = AUTHOR_EMAIL;

  message.addRecipient("Admin1", RECIPIENT_EMAIL_1);
  message.addRecipient("Admin2", RECIPIENT_EMAIL_2);

  String html;

  if (locked) {
    message.subject = "ALERT: Intruder Detected";
    html = "<h2 style='color:red;'>Security Alert!</h2>";
    html += "<p>System Locked due to failed attempts.</p>";
  } else {
    message.subject = "System Disarmed";
    html = "<h2 style='color:green;'>System Unlocked</h2>";
  }

  html += "<p><b>Live Stream:</b><br>";
```

```
  html += "<a href='" + streamUrl + "'>" + streamUrl + "</a></p>";

  message.html.content = html.c_str();

  if (!smtp.connect(&session)) {
    Serial.println("SMTP Failed");
    return;
  }

  if (MailClient.sendMail(&smtp, &message)) {
    Serial.println("Email Sent");
  } else {
    Serial.println("Email Failed");
  }

  smtp.closeSession();
}

// ============================
// SETUP
// ============================
void setup() {
  Serial.begin(115200);

  pinMode(TRIGGER_PIN, INPUT);  // ☑ FIXED

  camera_config_t config;
  config.ledc_channel = LEDC_CHANNEL_0;
  config.ledc_timer = LEDC_TIMER_0;

  config.pin_d0 = Y2_GPIO_NUM;
  config.pin_d1 = Y3_GPIO_NUM;
  config.pin_d2 = Y4_GPIO_NUM;
  config.pin_d3 = Y5_GPIO_NUM;
  config.pin_d4 = Y6_GPIO_NUM;
  config.pin_d5 = Y7_GPIO_NUM;
  config.pin_d6 = Y8_GPIO_NUM;
  config.pin_d7 = Y9_GPIO_NUM;

  config.pin_xclk = XCLK_GPIO_NUM;
  config.pin_pclk = PCLK_GPIO_NUM;
  config.pin_vsync = VSYNC_GPIO_NUM;
  config.pin_href = HREF_GPIO_NUM;

  config.pin_sscb_sda = SIOD_GPIO_NUM;
  config.pin_sscb_scl = SIOC_GPIO_NUM;
```

```
  config.pin_pwdn = PWDN_GPIO_NUM;
  config.pin_reset = RESET_GPIO_NUM;

  config.xclk_freq_hz = 20000000;
  config.pixel_format = PIXFORMAT_JPEG;

  config.frame_size = FRAMESIZE_QVGA;
  config.jpeg_quality = 12;
  config.fb_count = 1;

  if (esp_camera_init(&config) != ESP_OK) {
    Serial.println("Camera Init Failed");
    return;
  }

  WiFi.begin(ssid, password);
  Serial.print("Connecting WiFi");

  int timeout = 20;
  while (WiFi.status() != WL_CONNECTED && timeout--) {
    delay(500);
    Serial.print(".");
  }

  if (WiFi.status() != WL_CONNECTED) {
    Serial.println("\nWiFi Failed");
    return;
  }

  Serial.println("\nWiFi Connected!");
  Serial.print("Stream Link: http://");
  Serial.println(WiFi.localIP());

  startCameraServer();
}

// ===========================
// LOOP
// ===========================
void loop() {

  int state = digitalRead(TRIGGER_PIN);
  String url = "http://" + WiFi.localIP().toString() + "/";

  // LOCK EVENT
```

```
  if (state == HIGH && !isLocked) {
    delay(50);

    if (digitalRead(TRIGGER_PIN) == HIGH) {
      isLocked = true;

      Serial.println("LOCKED");
      sendStatusEmail(true, url);
      delay(2000);
    }
  }

  // UNLOCK EVENT
  else if (state == LOW && isLocked) {
    delay(50);

    if (digitalRead(TRIGGER_PIN) == LOW) {
      isLocked = false;

      Serial.println("UNLOCKED");
      sendStatusEmail(false, url);
      delay(2000);
    }
  }

  delay(10);
}
```